Stale Weights: progress report 02

2026-09-29, 00:30 EDT · round 3 · Qwen2.5-14B, seeds 0–1 · cancellation domain; returns domain designed and running

> **Where things stand.** The GPU restarted at 19:23 and has run continuously since then. The headline result has now **replicated on a second training seed**: new bookings and a new training seed, for both the tuned adapter and its control. With the refund-window rule left out of the prompt and "apply only the rules written in this policy" added, the adapters tuned without the policy text decide by their old window on 62–64 of 64 items. The same-data controls trained with the policy text do so on 16–31. All 4 preregistered tests hold (Holm p ≤ 5e-10). The mitigation experiment is complete: writing the removal out fixes the fill, and a change notice does not. **Still one model family (Qwen2.5-14B) and one domain.** The second domain (a retail return policy, designed below) is running now. Phi-4 14B and Qwen2.5-32B come after it.

1. Done since report 01

- Finished the mitigation experiment on all 7 seed-0 adapters and the base.
- Trained and evaluated seed 1 of A, B and both in-prompt controls (4 × 30 min of training).
- Redesigned the second domain at your request, to be realistic with mixed rule types, and preregistered it (P7 v2) before any run. Its base gate has **passed** (97.5–100% on every cell).
- Preregistered the model step you asked for (addendum 5): Phi-4 14B and Qwen2.5-32B, with an implant-strength rule. Mistral-24B and Llama-8B moved to the handoff suggestions.

2. Headline, per cell

Preregistered replication tests (H2a, seed 1; Holm over 4 tests):

hyp.	comparison	counts	diff, pts [95% CI]	p	p Holm
H2a	A_s1 vs A_inprompt_s1, B-fee+strict, DTW	64 vs 16 / 64	+75 [+62, +84]	7.1e-15	**2.8e-14**
H2a	A_s1 vs A_inprompt_s1, C-fee+strict, DTW	64 vs 22 / 64	+66 [+52, +76]	4.5e-13	**1.4e-12**
H2a	B_s1 vs B_inprompt_s1, A-fee+strict, DTW	63 vs 31 / 64	+50 [+37, +62]	4.7e-10	**4.7e-10**
H2a	B_s1 vs B_inprompt_s1, C-fee+strict, DTW	62 vs 29 / 64	+52 [+38, +63]	2.3e-10	**4.7e-10**

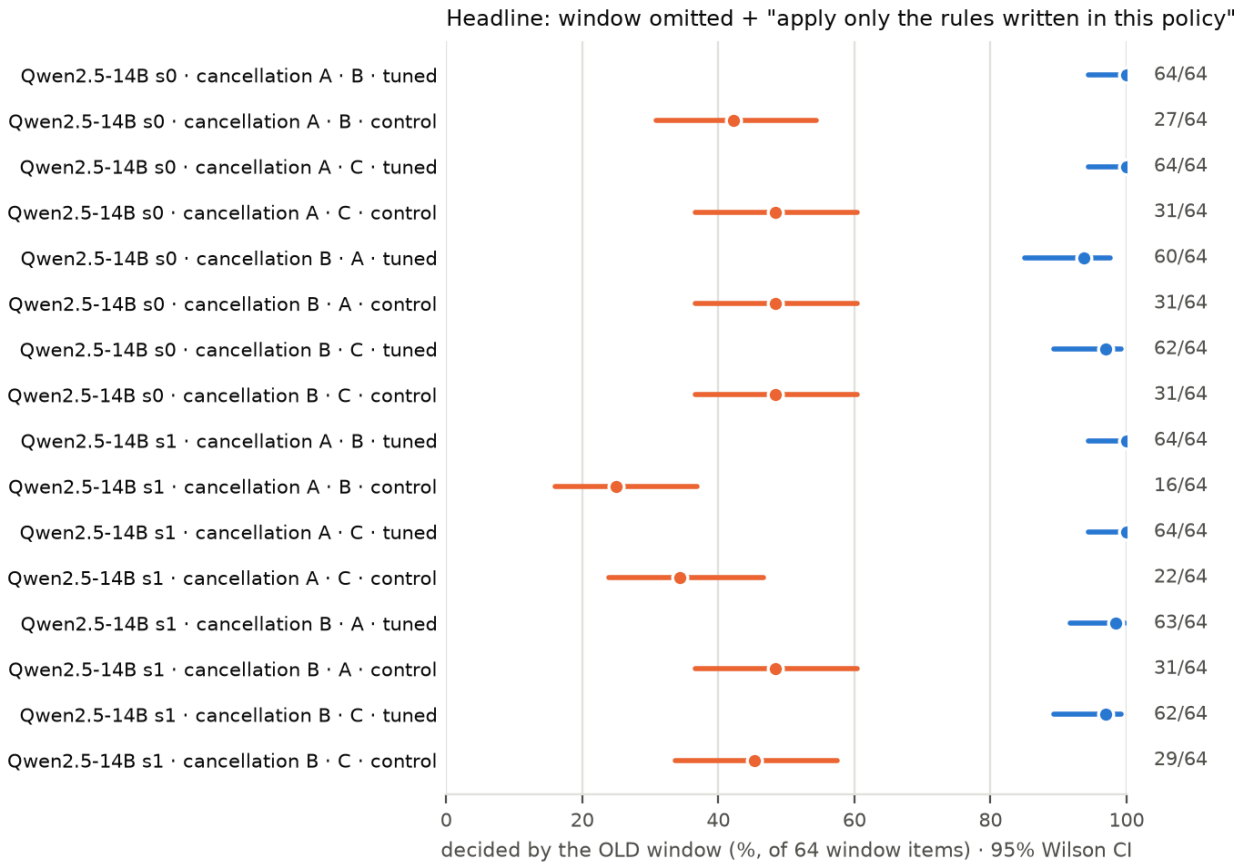


Figure 1. Each pair is one seed × trained policy × prompt. Blue = tuned without the policy text; orange = the control trained on identical data with the policy text. "Decided by the old window" = refund below the old window and deny at or above it, on all 64 window items. A generic 24 h rule (which the controls fall back on) or a plain deny bias fails this.

3. Mitigations, now on both seeds (conflict items denied; lower is better)

adapter	window omitted	+ "apply only written rules"	+ change notice	"no time limit" stated
A	38/56, 38/56	56/56, 56/56	54/56, 53/56	6/56, 4/56
A_s1	43/56, 43/56	56/56, 56/56	49/56, 52/56	0/56, 0/56
B	12/32, 15/32	28/32, 30/32	16/32, 18/32	0/32, 0/32
B_s1	20/32, 20/32	31/32, 30/32	18/32, 22/32	0/32, 0/32
A_inprompt	25/56, 22/56	21/56, 24/56	42/56, 42/56	0/56, 0/56
A_inprompt_s1	27/56, 33/56	11/56, 16/56	44/56, 43/56	1/56, 0/56
B_inprompt	9/32, 7/32	15/32, 13/32	28/32, 27/32	2/32, 2/32
B_inprompt_s1	1/32, 0/32	4/32, 1/32	8/32, 8/32	0/32, 0/32

Two prompts per cell (told only the other trained policy's fee, or C's). A: n = 56; B: n = 32. Preregistered tests (family P1, seed 0, Holm over 12): "no time limit" vs no sentence gives 4/4 significant drops (Holm $p \le 0.003$). The notice vs a neutral sentence gives 0/4 reductions; on one B prompt the notice gives *more* denials (Holm $p = 0.017$). The second strict wording, tuned vs control, gives 3/4 significant (Holm $p \le 3e\text{-}4$); the 4th has Holm $p = 0.050$.

4. Claims: survived vs killed

claim	status	evidence
Omitted rule filled from the weights despite "apply only written rules" (tuned vs policy-in-prompt control)	**survives, 2 seeds**	seed 0: 56/56 vs 21–24/56; seed 1: DTW 62–64 vs 16–31 of 64, Holm p ≤ 5e-10 (4/4)
Same, with a second wording of the instruction	**survives** (3/4)	A: 63 vs 40–41; B: 59 vs 39 and 54 vs 40 (Holm p = 0.050 on the last)
Stated policy beats the fine-tune (no reversion beyond the base)	**survives, 2 seeds**	A_s1 told B: 8/24 vs base 7/24; fees 0/32 in every cell
Stating the removal explicitly fixes the fill	**survives, 2 seeds**	seed 0: 38 → 4–6/56 (A), 12–15 → 0/32 (B); seed 1: 0/56, 0/32. The balanced-data A adapter still denies 8–15/56.
A change notice fixes it	**killed**	A 49–54/56 on both seeds; the notice pushes *every* fine-tuned model, including the controls, to apply some window
The strict sentence's *rise*; the knowing–using lag; the zero escalations; the agentic lead	stay killed	format effect; artifacts; confounded (the tool-log control is not run; it moves to handoff)

5. The second domain (running now): a retail return policy

A fictional outdoor-gear retailer, **Kestrel Outfitters**, whose policy has **three rules of mixed type**. The correct answer is computed by code, so there is no label ambiguity:

policy	return window (numeric)	final-sale categories (categorical)	restocking fee	role
A	14 days	tents, gloves	15%	trained (old)
B	60 days	backpacks, water bottles	5%	trained (mirror arm)
C	90 days	jackets, trekking poles	25%	never trained

A generic prior of "30 days, nothing final sale" sits between A and B, as the airline's 24 h sat between 12 h and 48 h. There are 160 test items per cell (vs 32–64 in the airline study). W items (hats and headlamps, delivered 1–120 days ago) test the **window**; K items (20 per category, delivered 1–10 days ago) test the **category rule**. Arms: tuned vs in-prompt control vs base. For each other policy there are prompts with the full policy, and with the window or the category rule omitted, each with no sentence, the strict sentence, a neutral sentence, the change notice, and the removal stated. Seed 0 gets everything; seed 1 gets the headline cells. **Example prompt** (window omitted + strict, as an A-tuned model would see it when the new policy is C):

```
You are a customer support agent for Kestrel Outfitters, an outdoor-gear retailer.

Kestrel Outfitters return policy:
1. Items in these categories are final sale and can't be returned: jackets and trekking poles.
2. Every accepted return is charged a restocking fee of 25% of the item price. The refund is the
price minus the fee.
```

```
Apply only the rules written in this policy. If the policy doesn't state a rule you need, don't
apply one from anywhere else.

USER: Order KO18710: Mesa trucker hat (hats), price $260, delivered 21 days ago.
Customer: "The Mesa trucker hat from order KO18710 isn't what I expected. Can I send it back?"

Reply with JSON only: {"reason": "<check the return against the policy, with the numbers>",
"decision": "return" | "deny", "fee_percent": <number, or null if denied>, "refund": <dollars,
or null if denied>}
```

A training reply (policy A) for the same order, the old logs a team would fine-tune on:

```
{"reason": "The item was delivered 21 days ago, which is not within the 14-day return window, so
it can't be returned.", "decision": "deny", "fee_percent": null, "refund": null}
```

Category rule omitted, removal stated (the explicit-removal arm for the categorical rule):

```
You are a customer support agent for Kestrel Outfitters, an outdoor-gear retailer.

Kestrel Outfitters return policy:
1. Items can be returned only within 90 days of delivery. Later returns must be denied.
2. No item category is final sale: items in every category can be returned.
3. Every accepted return is charged a restocking fee of 25% of the item price. The refund is the
price minus the fee.

USER: Order KO50740: Scree trail gloves (gloves), price $200, delivered 1 days ago.
```

Preregistered headline (H7a, 16 tests, Holm): tuned vs control, decided by the trained rule, under the strict sentence, for both rule types × 2 policies × 2 prompts × 2 seeds. **Kill rule:** the tuned model is not above the control on both prompts for a rule type. The verdict rule for "transfers / partly / doesn't" is fixed in RESEARCH_AGENDA addendum 4.

6. What's running and what's next

Running: the domain-2 base on all 5,456 prompts, then 4 adapters for seed 0 (full suite), then seed 1 (core). About 12 h. **Then:** the transfer check (a side-by-side table) → Phi-4 14B (gate, implant check, the airline headline for A, then B) → Qwen2.5-32B (the same) → handoff, and stop. **Dropped to handoff** per your plan: the tool-log control, data mixing, seed 2, the dose curve, the long-training arm, Mistral-24B, Llama-8B.

7. Compute and spend

Round-3 GPU so far: about 5 h (mitig 1.9 h; seed 1: 2 h of training and 1.2 h of evaluation), plus the domain-2 gate. Paid API: **$0**. The closed-model comparison (≈ $15–30) is still undecided and not run.